

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No Cummins® service tools required.

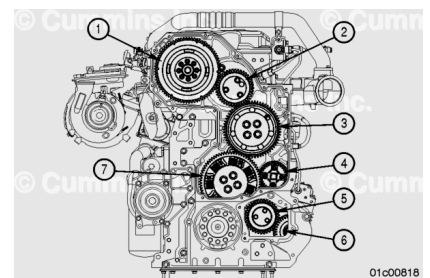
Additional Service Items

- Dial indicator

General Information

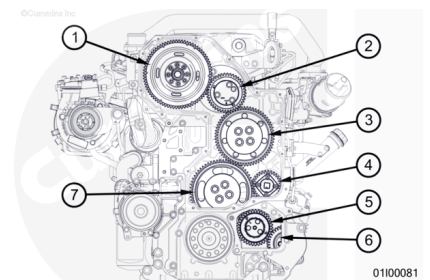
This illustration depicts the gear housing with the gear covers removed for gear trains with a scissor-style lower idler gear.

1. Camshaft gear, valve
2. Idler gear, adjustable
3. Idler gear, upper
4. Accessory gear
5. Idler gear, fuel pump, non-adjustable
6. Fuel pump gear
7. Idler gear, lower scissor.



This illustration depicts the gear housing with the gear covers removed for gear trains with a solid-style lower idler gear.

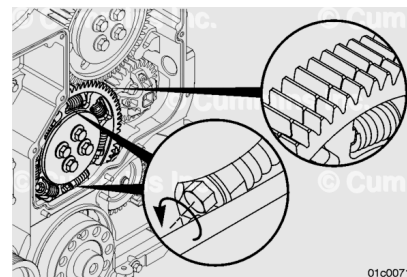
1. Camshaft gear, valve
2. Idler gear, adjustable
3. Idler gear, upper
4. Accessory gear
5. Idler gear, fuel pump, adjustable
6. Fuel pump gear
7. Idler gear, lower solid.



Measure

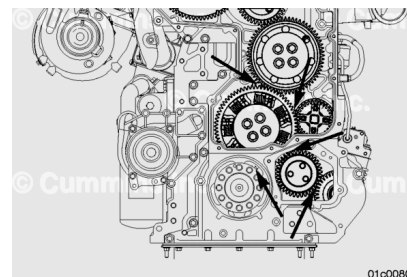
Unload the scissor-style lower idler gear, if equipped. Refer to Procedure 001-088 in Section 1.

(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)



Measure backlash for the following gear meshes between the two non-adjustable gears:

- Upper idler gear to lower idler scissors gear or upper idler gear to lower idler solid gear
- Lower idler scissors gear to fuel pump idler gear (non-adjustable)
- Lower idler scissors gear to accessory drive gear or lower idler solid gear to accessory drive gear
- Fuel pump gear to fuel pump idler gear (non-adjustable)
- Lower idler scissors gear to crankshaft gear or lower idler solid gear to crankshaft gear.



⚠ CAUTION ⚠

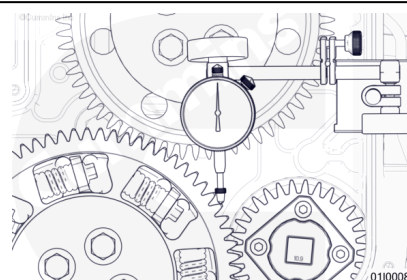
The adjacent (engaging) gear must not turn as the gear being measured is turned.

Position a dial indicator so the tip is contacting the surface of the gear tooth, as illustrated. Do **not** allow the mating gear to turn.

Rotate the gear being measured **clockwise**.

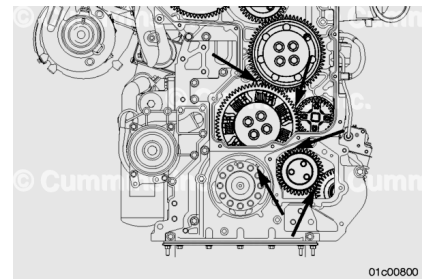
Set the indicator to "0".

Rotate the gear **counterclockwise** and read the indicator.



Gear Train Gear Backlash		
mm		in
0.0508	MIN	0.002
0.46	MAX	0.018

If the backlash measurements for the following are out of range, adjust the backlash. Refer to Procedure 001-088 in Section 1. (/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)



Repeat the backlash measurements. If the backlash measurements are still out of range, remove the gears and inspect for signs of damage. If damage is found or if the backlash is still out of specification, the gear(s) **must** be replaced. Refer to Procedure 001-088 in Section 1. (/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)

Measure the backlash for the following gears:

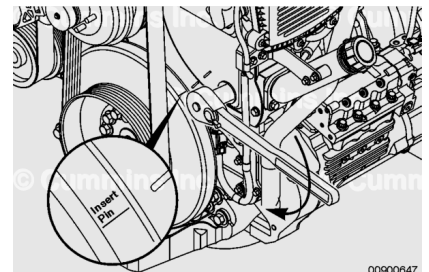
- Upper idler gear to lower idler scissors gear or upper idler gear to lower idler solid gear
- Lower idler scissors gear to fuel pump idler gear (non-adjustable)
- Lower idler scissors gear to accessory drive gear or lower idler solid gear to accessory drive gear
- Fuel pump gear to fuel pump idler gear (non-adjustable)
- Lower idler scissors gear to crankshaft gear or lower idler solid gear to crankshaft gear.

Measure backlash for the following gear:

- Camshaft idler gear (adjustable) to camshaft gear.

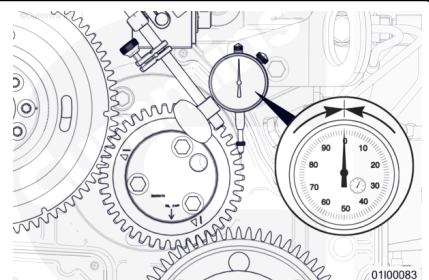
Rotate the engine **clockwise** so there is a space between the mating teeth of the camshaft gear and the adjustable idler gear.

Insert the largest possible feeler gauge in the gear mesh location indicated by the line on the adjustable idler gear retainer.



Position a dial indicator so the tip is contacting the surface of the tooth as illustrated.

Set the indicator to "0".



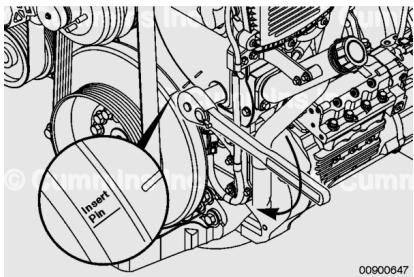
⚠ CAUTION ⚠

The adjustable idler gear must not rotate.

Gently rotate the engine **counterclockwise**. **Only** rotate the engine enough to take up the gear lash. Do **not** bar over the engine.

Record the dial indicator value.

Add the values of the feeler gauge and the reading of the dial indicator.



Gear Backlash (feeler gauge values plus indicator value)

mm		in
0.102	MIN	0.004
0.3048	MAX	0.012

The value of the feeler gauge or the dial indicator **must not** exceed 0.20 mm [0.008 in].

Use the following procedure to adjust backlash if the backlash does **not** meet the above specifications. Refer to Procedure 001-088 in Section 1.

(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)

Repeat the backlash measurement after the adjustment.

Note : The camshaft gear **must** be removed, the camshaft nose cleaned, and the camshaft gear installed while adjusting the backlash on the camshaft idler gear. See Install section for details. Refer to Procedure 001-088 in Section 1.

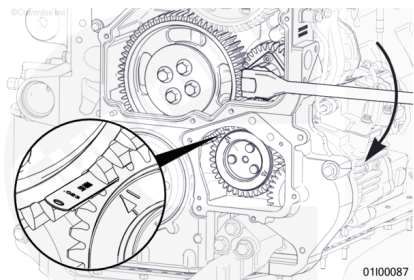
(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)

Measure backlash for the following gear:

- Fuel pump idler gear (adjustable) to lower idler gear.

Rotate the engine **clockwise** so there is a space between the mating teeth of the lower idler gear and the fuel pump idler gear.

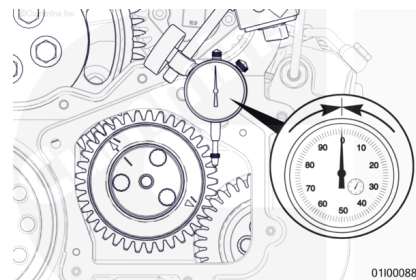
Insert the largest possible feeler gauge in the gear mesh location indicated by the line on the adjustable idler gear retainer.



Position a dial indicator so the tip is contacting the surface of the tooth as illustrated.

Set the indicator to "0".

Note : If a dial indicator can not be properly set up on the fuel pump idler gear, the use of **only** the feeler gauges is acceptable.



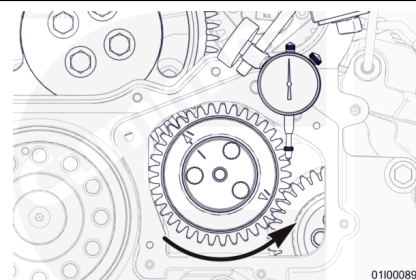
⚠ CAUTION ⚠

The fuel pump idler gear must not rotate.

Gently rotate the engine **counterclockwise**. **Only** rotate the engine enough to take up the gear lash. Do **not** bar over the engine.

Record the dial indicator value.

Add the values of the feeler gauge and the reading of the dial indicator.



Gear Backlash (feeler gauge values plus indicator value)

mm		in
0.102	MIN	0.004
0.3048	MAX	0.012

Use the following procedure to adjust backlash if the backlash does **not** meet the above specifications. Refer to Procedure 001-088 in Section 1.

(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)

Repeat the backlash measurement after the adjustment.